From Imitation Collapse to Visitation-Corrected Distillation: Training a 2B-Parameter MMORPG Tool-Use Agent

A three-round on-policy distillation case study (Kaetram)

Kaetram Agent Project

Technical report, revised July 18, 2026 — single-codebase case study

Abstract

We report an end-to-end case study of post-training a small language model (Qwen3.5-2B) to play Kaetram, an open-source MMORPG, through a typed tool-call interface (17 MCP tools, OODA loop, 16K context). **The headline: on-policy distillation makes the 2B better than it started** — it clears a quest wall the base model never passes and finishes at **18/30**, up from base's 12, completing two of the three quest chains unseeded where base completes one. The cleanest piece is weights-only: on the same harness, the distilled student passes that wall 3/3 where base passes it 0/3. What follows is the honest accounting of which lever did what. Across eleven training rounds we traversed three regimes. (1) **Cross-vocabulary SFT on frontier-teacher trajectories regresses the student** (3.5× drop on our Core-3 quest benchmark; Mann-Whitney exact p=0.029): uniform cross-entropy imprints the corpus's *action marginal* rather than its policy. (2) **Harness/state-contract engineering dominates capacity**: a scaffold worth +12 benchmark stages (base-9B 7 → 19 on matched full runs) transfers from 9B down to 2B, while scaling to 27B adds nothing. (3) **Reverse-KL on-policy distillation** (4B teacher → 2B student) transfers *style without competence* in one round — the student exactly matches the teacher's tempo and tool mix yet gains zero task stages — and we show why: the loss is an expectation over the *student's* visitation distribution, which never reaches the states where the teacher's competence lives. Round two corrects visitation by **seeding the environment database** at the failure state (a reverse-curriculum state reset, here a write of persistent typed environment state for distillation rather than reward-RL), and the student passes the wall it had never passed: 12/30 → 15/30, with all three eval agents clearing the gating quest stage unseeded (0/3, 0/3, 3/3 across arms). Round three reaches the program-best **18/30** by breaking the next wall, but a controlled ablation attributes most of that round-two-to-three step to a harness recovery affordance — the round-three weight update's reliable payoff is ∼2× faster progression, not stages — so we label 18/30 a weights-plus-interface result and do not lean on nominally edging the 4B teacher (18 vs 17, with a recovery fix it lacked). We document a failure mode we believe is novel: under teacher forcing the **teacher prefers the student's malformed tool syntax** (≈ 85% probability on a continuation it never emits generatively) — an in-context copy prior surfacing inside the grader — which dense reverse-KL distills. Suppressing the teacher's endorsement of the defect in the grading context does not suppress the student's generation of it; a small harness recovery affordance with in-context correction does (98.5% self-correction, zero relapse). The program's ceiling is over-determined: Rick's Roll stays 0/4 gated at link 1 by a displaced fishing tool (cooking is never reached), behind which the cook-incompetent same-family teacher transfers *zero* cook behavior — a teacher cannot grade in what it cannot do, one of several blockers. Every arm is n=1 across three personality-variant agents; we report the limits alongside the results.

1 Introduction

Can post-training instill genuinely long-horizon, tool-mediated competence in a ∼2B-parameter model — as opposed to surface style? This report traces a single project's complete answer-so-far across three training paradigms, eleven rounds, and one environment, with every claim tied to logged rollouts and code.

The task: autonomously progress three quest chains ("Core 3", 10 stages total) in Kaetram, a 2D MMORPG, via structured tool calls only. The metric: new quest stages per 6-hour run, summed over three personality-variant agents (*stages*/30). The arc:

1. **r1–r9 (SFT iteration).** Nine rounds of supervised fine-tuning of Qwen3.5-9B on Claude-teacher trajectories established methodology (loss masking, prompt parity, truncation gates, a $\sqrt{r}$ LoRA trap) but never beat the base model.

2. **r10 (the clean negative).** A rigorously controlled SFT round *regressed* the student 3.5× below base with a measurable mechanism: the trained policy's tool-call distribution matches the *training-corpus marginal* within ∼1pp on decision verbs — corpus prior becomes inference prior.

3. **r11a (the scaffold reframe).** Harness and state-contract engineering — not prompt wording, not parameters — moved base models from 4/30 to 19/30. The scaffold transfers across model sizes (27B/9B/4B/2B), the walls are identical at every size, and 27B buys nothing: capacity is not the lever.

4. **r11b (this study's focus): three rounds of on-policy distillation**, scaffolded Qwen3.5-4B (17/30) → base Qwen3.5-2B (12/30). Round one transferred the teacher's *behavioral distribution* with striking precision and zero task gain. Round two, after correcting the visitation distribution with environment-state seeding, produced the first weights-driven benchmark lift: 15/30, unanimous unseeded passage of the previously-unpassed quest wall. Round three reached 18/30 — the project best — by breaking the next wall, but required a *harness* affordance to fix a format defect that weights-side fixes could not, and hit a hard ceiling on Rick's Roll (0/4) where the same-family teacher is itself incompetent. The trajectory is monotone (12 → 12 → 15 → 18) but each gain came from a *different* lever, and we are explicit about which.

Contributions. (i) A documented instance of *style-without-competence* transfer under reverse-KL OPD, with a per-verb KL×behavior cross-table showing sign-consistent coupling, including a double-sided mode-seeking overshoot. (ii) **Environment-state seeding for agentic distillation**: shifting the student's visitation distribution d^{π_S} by seeding the game database at teacher-demonstrated failure states, while leaving the loss untouched. This is a reverse-curriculum state reset — the RL lineage [19, 20, 21] and the LLM text-prefix analogues [18, 24] are the relatives; our instantiation (a write of persistent typed environment state, for on-policy distillation not reward-RL) is the new part, causally validated here (seed → wall-state data → unseeded passage). (iii) The **teacher-forcing copy-prior** failure mode: a generatively-clean teacher that assigns ≈85% to continuing the student's malformed tool syntax in-context, which dense OPD therefore distills; we measure it, mask it, and track its mutation across rounds, and show that *grading-context* fixes (counterfactual canonicalization), though they verifiably suppress the teacher's endorsement, do not transfer to the student's *generation* — only a harness recovery affordance with in-context correction cures it. (iv) A **clean separation of levers**, quantified by a controlled ablation: over base, the *weights* instill the competence (the wall passes 3/3 unseeded, a weights-only result); an interface affordance dissolves a defect no weights update reaches (it carries most of the round-two-to-three *step*, while the round-three weight update buys speed); and a same-family teacher transfers *zero* of a behavior it never performs (cook = 0) — one of the over-determined walls behind Rick's 0/4, where a displaced tool already gates link 1. The harness+weights co-evolution thesis, made concrete.

2 System

2.1 Environment and benchmark

Kaetram-Open is a tile-based MMORPG: a Node game server per agent (WebSocket), MongoDB persistence, browser client. **Core 3** totals 10 stages: *Foresting* (3 stages; two 10-log deliveries), *Herbalist's Desperation* (3: accept → 3 bluelily → 2 paprika + 2 tomato; herbs require Foraging level 5), *Rick's Roll* (4: talk → 5 cooked shrimp → a door puzzle → delivery). Two mechanics make this hard for an LLM agent: *skill gating* (resources refuse interaction below the required level) and *stochastic yield* — each server tick a harvest succeeds with low probability (≈1/3 for an unleveled player, rising with skill level), and success depletes the bush. Long-horizon, partially random gathering — not a scripted sequence. The metric, CORE-3, is new stages reached: per agent-run the first-to-last observed stage delta (so resume-state replay cannot inflate it), summed over the three agents, in $[0, 30]$.

2.2 Agent harness

The agent runs an OODA loop (`play_qwen.py`): an OpenAI-compatible client against a Modal SGLang endpoint plus an MCP stdio client driving a Playwright-controlled browser. Decoding uses Qwen3.5's thinking-mode sampling preset (temp 1.0, top-p 0.95, top-k 20, presence 1.5), but the generation prompt is the template's *non-thinking* default — a closed-empty `<think></think>` block, so reasoning surfaces as plain content (the dangling-`</think>` dialect, §5.3), a regime held identical across base, teacher, and student. Context budget 14,336 of the 16,384 training length, measured with the same patched chat template the server renders; overflow triggers a *session rollover* (fresh conversation, persistent game state, a programmatic — never model-authored — advisory note carrying quest stage and position). The MCP server exposes 17 tools with three design principles: *enforced affordances* (attack auto-loots; navigate does BFS with snap-to-walkable; craft/buy auto-walk), *advisory state-contract* (`query_quest` leads with canonical facts — accepted/stage/needed/have/remaining — plus a recommendation the agent must verify; `observe` enriches mobs with level/aggression and quests with item progress), and *payload compaction* (ASCII map dropped; ∼2× turns per session). Three agents run in parallel (personality variants: grinder / completionist / explorer — priority modifiers only), each with its own game server, sandbox, and display; `orchestrate.py` supervises; runs are launched from a wiped, freshly-seeded database.

2.3 Training and serving infrastructure

Each model size is a Modal SGLang app (student L4, teacher A100) exposing `/v1/chat/completions` for rollouts and `/v1/score` for teacher-forced per-token logprobs over a caller-supplied (context, full-text) pair — raw pre-rendered strings, because the Qwen3.5 repos ship different chat-template revisions but share one vocabulary, so an identical string tokenizes identically on student and teacher. All components apply one chat-template patch (the stock template silently strips `<think>` from intermediate turns; QwenLM/Qwen3 #1831). Training runs on a Modal H100 with Unsloth +

fresh LoRA (r=64, α=64, rsLoRA off); Qwen3.5's 3:1 Gated-DeltaNet hybrid requires pinned fast kernels (`triton==3.3.1` + `fla 0.5.0`), which we adopted only after a three-way fp32 parity probe (fast-bf16 vs fallback-bf16 vs fp32 reference on real corpus tokens; the fast path proved marginally *closer* to fp32 than the fallback, grad cosines $\geq$0.997, 8.9$\times$ speedup — round-2 training: 50 minutes vs round-1's 8 hours).

3 The SFT era (r1–r10): imitation collapse

Rounds r1–r9 (Qwen3.5-9B, Claude-trajectory SFT) produced methodology, not wins: completion-only loss masking (and the discovery that three rounds had silently trained full-token); chat-template CoT preservation; byte-identical train/eval prompt parity; strict 16K truncation gates; and the rsLoRA trap (α=r under rsLoRA is an 8$\times$ effective-LR error that diverged r7). r9 still lost to base.

r10 made the comparison airtight and the result negative. Corpus: 8,510 train records from 15 archetype-diverse Claude runs, quality-gated. Result: base scored 7/30 in *all four* eval runs (zero variance); r10-SFT scored $\{3, 1, 2\}$, mean 2.0/30 — a 3.5$\times$ regression (Mann-Whitney exact, one-sided, p=0.029; Fisher's exact on Foresting completion p=0.016, OR= 16). The mechanism is legible in the tool mix: at inference the SFT policy's decision-verb frequencies match the *training-target* distribution within $\sim$1pp (`interact_npc` corpus 2.4% $\leftrightarrow$ SFT 2.1% vs base 10.8%; `query_quest` 2.2 $\leftrightarrow$ 2.5 vs 11.2; `navigate` 27.7 $\leftrightarrow$ 26.4 vs 5.5). Credit analysis showed why that marginal is poison: 52% of source sessions advance no quest; `interact_npc` carries the highest near-term credit (36% of uses precede an advance within 5 turns) yet is 3.8% of the corpus, while `attack` is 25.5% of corpus with $\approx$0% credit. Uniform cross-entropy imitated the wrong marginal — *corpus prior becomes inference prior.* Diagnosis mapped to four known mechanisms: compounding off-policy error (DAgger's $O(\epsilon T^2)$), capacity-gap marginal-fitting, CE diversity collapse, and capability suppression.

4 The scaffold reframe (r11a): the harness is the lever

Before more training, we asked what a *frozen* model could do with a better model–environment contract. Over eight days the base-9B Core-3 best-run envelope rose through 4, 9–10, 12, 15, and 19/30 — every causal jump attributable to harness/state-contract changes (quest-list normalization, canonical `current_step` facts, navigation hardening), none to prompt wording (four survival-rule rewrites moved nothing). The 4 is the short, pre-`current_step` development floor; on matched full runs the clean baseline is 7 (Table 1), so the controlled scaffold lift is 7 $\rightarrow$ 19 (+12). The knowing-doing gap localizes at the boundary between advisory and enforced behavior: every affordance that works is *enforced* (auto-loot, BFS navigate); every remaining blocker is *advisory* (survival rules the model quotes and ignores).

The scaffold transfers across capacity. With identical scaffold and prompts: 27B = 15/30 (n=1, $\sim$4h), 9B = 12–19/30, 4B = 12/30 (3h) and **17/30** (5h), 2B = 13/30 (5h) and 12/30 (6h). Foresting is solved at every size and Rick's Roll is essentially untouched at every size (only the 9B's best run reaches Rick's stage 1); the *Herbalist wall* is where capacity shows — the 2B stalls at stage 1 while 4B, 9B, and 27B all clear it (the 27B also gained a tool-result confabulation failure mode). Above the 2B, capacity barely moves the score — the 27B's 15 sits inside the 9B's 12–19, so scaling up buys nothing — but the gap that matters is the 4B's 17 vs the 2B's 12, which sets up the distillation experiment: **teacher = scaffolded 4B, student = base 2B** — a real, same-family competence gap, an order of magnitude cheaper per round than the 9B lane. (Reference ceiling, on the pre-R11 Claude Code harness: in its best run 2/3 of the Claude agents complete *all* of Core 3 including

Table 1: **Program-wide comparison — quest progression and scale.** Every run in this report, grouped by harness era (the central variable). Quest columns count agents (of 3) completing each chain; Core-3 is new stages reached (last−first observe) summed over the three agents. The R11 block is the controlled comparison (identical scaffold, varying capacity); pre-R11 rows are different-harness reference points (Claude on the Claude Code CLI). Error rate is the game-state failure rate — the high-error rows (Claude, OPD-r3) are attempting the hardest content (Rick's door; the hostile Herbalist zone), not failing the basics. Durations differ (3–6h) so tool calls/deaths are not duration-normalized; Core-3 (a first-to-last delta) is the duration-robust metric.

Model	Core-3/30	Forest.	Herb.	Rick's	dur	calls	err%	deaths
pre-R11 harness — different from the R11 Qwen ladder								
Claude Sonnet	27	3/3	3/3	2/3	5.9h	5,605	12.9	114
base-9B (r10-era)	7	2/3	0/3	0/3	3h	1,738	5.1	36
r10-SFT 9B	2	0/3	0/3	0/3	3h	1,610	13.3	62
R11 scaffold — controlled: identical harness, varying capacity								
27B	15	3/3	1/3	0/3	4.2h	2,324	3.4	146
base-9B	19	3/3	3/3	0/3	5h	2,844	3.4	145
4B (teacher)	17	3/3	2/3	0/3	5h	2,140	4.3	105
base-2B (student)	12	3/3	0/3	0/3	6h	6,183	8.6	142
OPD-r3 2B (+recovery)	**18**	3/3	3/3	0/3	6h	8,563	13.0	183

Rick's Roll — 27/30, Table 1 — in 6h at 1,400–2,300 tool calls. This shows the game is winnable, not a same-harness ceiling for the Qwen ladder.)

Program-wide comparison. Tables 1 (quest progression & scale) and 2 (tool-use signature) collect every run in this report, grouped by harness era. They make three things visible at once: the *harness lift* (the same base-9B goes 7 → 19 from the pre-R11 to the R11 harness), the *SFT regression* (r10-era base 7 → r10-SFT 2, with the tell-tale navigate amplification and decision-verb suppression in Table 2), and the *capacity ladder* at fixed R11 harness (2B 12 / 4B 17 / 9B 19 / 27B 15, saturating above 9B) into which OPD lifts the 2B (12 → 18). Only the R11 block is a controlled comparison; the pre-R11 rows (Claude, r10) are different-harness reference points.

5 On-policy distillation: method

5.1 Objective and loss

Let π_S, π_T be student and teacher, and d^{π_S} the state distribution induced by student rollouts (a "state" is the full serving prompt: system, bootstrap, dialogue history). Batched reverse-KL OPD minimizes

$$\mathcal{L}_{\text{OPD}} \;=\; \mathbb{E}_{s\sim d^{\pi_S}}\; \mathbb{E}_{a\sim\pi_S(\cdot|s)}\Big[\;\log\pi_S(a\mid s)\;-\;\log\pi_T(a\mid s)\Big], \tag{1}$$

the I-projection (mode-seeking) direction, estimated on the student's own emitted tokens with the teacher as a per-token grader. We freeze a per-token advantage at data-build time — the negated student-minus-teacher logprob gap, taken against the *behavior* (rollout-time) logprobs — and train with PPO-style clipped importance sampling (ϵ=0.3, advantage clamp ±3, plus a 1.5× weight on each session's first third). The clip is load-bearing: unclipped REINFORCE on a fixed batch's majority-negative advantages pays the optimizer to drive those logprobs to $-\infty$ and degenerates the policy to an `observe` loop — a collapse we hit and the clip prevents. Because each round's init equals the rollout-generating policy (fresh LoRA over the previous round's merged checkpoint), the

Table 2: **Program-wide comparison — tool-use signature** (% of tool calls; same rows and grouping as Table 1). The r10-SFT collapse is legible: `navigate` amplified to 27.9% (from base-9B's 5.2%) while the decision verbs (`query_quest`, `interact_npc`) are suppressed — the corpus-marginal imprint. The R11 models lean on `query_quest` and `gather` (the scaffold's progression verbs). †Claude's turns/session is `maxSessionTurns`-bounded, not comparable to the Qwen agents' 16K-context rollover sessions; use *tool calls* (Table 1) as the cross-harness scale.

Model	observe	navig.	attack	gather	q_quest	npc	turns/s
pre-R11 harness							
Claude Sonnet	28.2	23.6	18.7	11.3	1.5	3.2	151†
base-9B (r10-era)	39.5	5.2	15.5	4.3	4.1	8.7	3.7
r10-SFT 9B	32.7	**27.9**	13.9	5.5	2.1	2.7	3.7
R11 scaffold							
27B	31.9	22.0	2.8	17.7	8.8	1.5	8.2
base-9B	36.5	13.6	10.0	12.9	11.4	2.1	6.1
4B (teacher)	36.1	9.1	9.1	20.8	8.7	1.4	6.1
base-2B	26.6	14.3	17.9	17.9	5.3	3.8	9.1
OPD-r3 2B	30.3	21.2	10.1	11.4	13.3	3.1	8.0

importance ratio is ≈ 1 at step 0 and the clip doubles as a KL trust region around behavior; a step-0 calibration probe verifies the serving-vs-training numeric gap (round 2: median $|\Delta \log p| = 0.0005$, clip fraction 0.000). LR 5×10^{-5}, one epoch, effective batch 32, H100; the LM head is applied only at action positions (a full-vocab forward at 16K $\times$ 248K would OOM).

5.2 Visitation coupling and environment seeding

Equation 1's expectation over d^{π_S} is the binding constraint this study makes concrete: the teacher is only ever *queried* at states the student visits. If the teacher's competence edge lives at states of measure ≈ 0 under d^{π_S} — e.g. standing at a Blue Lily Bush with Foraging $\geq$5 — no amount of dense grading transfers it. Round 1 demonstrates the failure; rounds 2–3 demonstrate the repair: change the *state distribution*, not the loss. Concretely we mix the student's natural visitation with rollouts of π_S launched from DB-seeded milestone states — player rows written directly into the game's MongoDB at e2e-verified mid-quest configurations (quest stage, skills, inventory, position), $\sim$30% of records by count. Crucially the loss, the teacher, and the on-policy character of the data are untouched — only the *state source* changes, which keeps the causal attribution clean. Resetting to hard or intermediate states to get learning signal there is a well-trodden idea in RL — reset along a demonstration [19], Backplay's reverse curriculum [20], Go-Explore's restore-then-explore [21], and expert-state reset mixtures for LLM agents [22]; the LLM-side analogues operate in text-prefix or policy space (branched expert anchors [18], behavior-policy blending [24]). Our narrow contribution is the *setting and mechanism*: a reverse-curriculum state reset instantiated as a *write of persistent, typed environment state* (a DB row), used for on-policy *distillation* rather than reward-RL — "Backplay for LLM tool-use distillation," which we have not seen done.

5.3 Data construction: the prefix property

Each action turn is rebuilt from rollout logs as the byte-exact serving prompt (system + bootstrap + last 28 messages rendered through the patched template with `add_generation_prompt`) plus the **verbatim logged emission** appended directly: full = ctx $\|$ emission. Re-rendering the completed turn through the chat template is wrong in a subtle way we hit twice: the template extracts reasoning from any `</think>` inside content and re-renders a real think block, diverging from the closed-empty-think generation prompt the model was actually served — 4% of base-2B turns in round 1, but 99% of r1-policy turns (whose acquired dialect ends content with a dangling `</think>`). Verbatim construction makes the prefix property hold by construction; the round-2 build of 7,849 states had *zero* alignment failures of any kind. Both endpoints score the same raw string; behavior logprobs are therefore exactly the rollout policy's serving distribution. Every 10th session is held out for the gate.

5.4 Abstention masking and the gate

Advantages are zeroed on token spans matching malformed parameter keys (`<parameter=K=V>`) where the teacher's grading is probe-verified unreliable (§6.2); 177 spans / 1,422 tokens (0.06%) in round 2. (Round 3 grades these same spans correctively instead of masking them, §8.) The post-train gate scores the new student on the held-out set and reports the token-weighted reverse-KL to the teacher over scorable action tokens (masked spans excluded). That number is *directional only*: round 1 used a $\geq$ 30% reduction as a PASS bar and it *failed* (+10.9%) while behavior moved dramatically — KAT [7] shows per-token KL on teacher-agreement-trapped spans is a misleading quality signal — so rounds 2–3 gate hard on behavioral checks instead: sampled-completion malformed-rate not worse than the prior round, no degeneration, and no catastrophic reverse-KL blow-up. Round-2 gate +14.5%, round-3 gate +6.4% (diminishing, as expected once the student is near the teacher), both PASS.

6 Round 1: style transfers, competence does not

Setup. Corpus: the 2B's own 6h baseline rollouts (5,564 train / 574 held-out records) re-scored by the 4B; the pre-train diagnostic put the largest student–teacher disagreement exactly on the 2B's documented weak verbs (eat_food +0.298 nats, respawn +0.292). Training: 174 steps. Eval: 6h, 3 agents, protocol-identical to baseline (compact observes, fresh DB, parity verified).

Result: Core-3 12/30**, identical to baseline** — every agent 4/10 with the same split, stuck at the same Herbalist stage-1 wall. Underneath, the student moved decisively onto the teacher's distribution:

	base-2B	2b-opd-r1	4B teacher
turns/session	9.1	**6.1**	6.1 (exact)
observe share	26.6%	**35.5%**	36.1%
navigate share	14.3%	**9.4%**	9.1%
attack share	17.9%	**6.6%**	9.1% (overshot)
query_quest share	5.3%	**18.0%**	8.7% (overshot 2×)
chars/turn	411	**1248**	864 (overshot)
game-state error rate	8.5%	**5.5%**	4.3%

Execution improved where the signal pointed (navigate errors 5.3%→1.2%; eat-inedible halved; phantom-target re-attacks replaced by re-grounding observes), with textbook mode-seeking signatures: exact convergence on the teacher's high-frequency modes and amplification *past* the teacher on its preferences.

6.1 The KL×behavior cross-table

The held-out gate moved only +10.9% (a formal FAIL against its 30% bar) while behavior moved dramatically — and the per-verb *signs* line up: every verb whose rKL dropped (attack −17%, stuck_reset −23%, warp, respawn, navigate) moved toward the teacher behaviorally, and `query_quest` — the single verb whose rKL *increased* (+20%) — is exactly the verb that behaviorally overshot the teacher 2×: the student moved *past* the teacher and diverged from the other side. Small per-token movement compounds over autoregressive rollouts; a token-KL gate on old-visitation states measures the wrong quantity.

What the advantages reward: structure, not reasoning. Reading the round-1 held-out advantages directly (teacher minus student logprob over the emitted tokens), the positive reverse-KL signal concentrates on tool-call *structural* tokens — the `</think>` delimiter (+0.212, 79% positive) and the `<function=`/`<parameter=` syntax (sharply positive) — while *reasoning-prose* tokens carry the most *negative* mean advantage (−0.333; the tool-call region as a whole averages ≈0). The `</think>` reward — the dialect's entry hook — explains the tag rate jumping 4% → 99%; the ∼4× growth in reasoning *length* is instead a trajectory-level mode-seeking pull toward the teacher's reason-then-act trajectories, *not* a per-token reward. So OPD transfers the *shape* of reasoning, not its substance — and the round-3 compression (§8) is the same signal with that pull relaxed.

6.2 The teacher-forcing copy prior

Round 1 introduced a defect: 8.3% of argument-requiring calls arrived with empty input, dominated by one span family — the model writes the boolean kwarg into the parameter *key*, `<parameter=accept_quest_offer=True>` (154 malformed vs 18 correct in the r1 eval; 0 vs 182 in baseline), which the parser drops, silently no-op-ing quest accepts. Three candidate mechanisms were refuted by direct measurement (the training records carried only the correct form; the teacher prefers the correct delimiter +0.74 nats in base-rollout contexts; no reasoning-priming association). The live probe settles it: **on the student's own malformed prefix, the 4B teacher assigns** −0.161 **nats (≈85%) to continuing the malformation** vs the student's −0.253 — advantage +0.09 *toward* the defect — while at correct-form states it grades +0.39 toward the correct token. The teacher's grading flips with context: an in-context copy/induction prior surfaces inside the grader under teacher forcing, even though the teacher never emits the form generatively. Dense reverse-KL therefore faithfully distills the defect and cannot self-correct it on the student's own rollouts. We find no precedent for this mechanism in the distillation literature (nearest relatives: the "KL agreement trap" [7], the survey's flawed-prefix trap [6], SKD's inaccurate teacher feedback [16]); the mechanistic substrate is the ICL copy bias / induction-head literature [28, 29].

7 Round 2: visitation-corrected distillation

Environment-state seeding. The 4B teacher passes the Herbalist wall in its own runs (2/3 agents finished the quest) — the gap is real and teacher-demonstrated — but base-2B rollouts essentially never visit lily-bush-at-Foraging≥5 states (20 gated attempts in 6,183 calls). We therefore seeded the *environment database*: player rows written mid-quest (Herbalist stage 1 accepted, Foraging XP 450 = level 5 on the game's XP curve, starter kit, spawn at the e2e-verified walkable tile beside the lily cluster), then a 2.5h collection run of the r1 policy from those states. This changes d^{π_S} only — the loss, the teacher, and the on-policy character of the data are untouched. In the seeded rollouts all three agents passed the wall (the first 2B passages ever observed), producing both student-authored success trajectories and dense teacher grades at decision states. This is a reverse-curriculum state reset — the RL lineage (reset-along-a-demo [19], Backplay [20], Go-Explore [21], expert-state resets [22]) and the LLM-side text-prefix/policy analogues (branched rollouts [18], behavior blending [24]); our instantiation, a write of *persistent typed environment state* for on-policy *distillation* (not reward-RL), is what we have not seen published.

Corpus and training. 7,849 states (5,523 natural r1-eval rollouts + 2,326 seeded wall states, interleaved) → 7,024 train / 825 held-out records, zero construction failures; fresh-state per-verb rKL came in at ≈half of round-1 levels (eat_food +0.298 → +0.149; Herbalist frontier +0.261 → +0.136) — round 1 had genuinely moved the policy toward the teacher on its own visitation. 220 steps, 50 minutes. Gate: **PASS** — held-out rKL 0.4281 → 0.3662 (+14.5%, *every* verb improved, including `query_quest` reversing its round-1 sign), no degeneration, no blow-up.

Eval (6h, unseeded, parity verified): Core-3 15/30 — 5/10 for every agent. All three agents passed the Herbalist stage-1 wall unseeded (at 4.1h, 4.5h, 5.0h) — a stage neither base nor r1 ever reached: arm-level passage rates 0/3, 0/3, 3/3. Supporting movement: all accepts landed by 3.2h (prior arms' stragglers: 5.7h, 5.3h); correct-form accepts 26 vs 86 malformed (r1: 18 vs 154); Foresting completion monotone faster across rounds (0.4–0.6h vs 0.5–0.8 vs 0.7–1.8); the round-1 overshoots retreated toward the teacher (query_quest 18.0 → 12.9%; verbosity 1248 → 1116 chars/turn), with turns/session holding at the teacher's tempo (6.1 → 6.0). Qualitatively, the passage mechanism is visible in the logs: r2 agents rotate bushes after empty harvests, track the Foraging-5 gate explicitly, and exploit gather's auto-walk; r1 at the same wall exhibited a *premature-turn-in attractor* (standing one tile from the bush holding 2/3 lilies, it walks 85 tiles to the quest NPC with an incomplete set) and gate-blind retries at Foraging 1–4; base displaced into combat (a level-59 character with Foraging 4). At run end all three r2 agents were mid-stage-2 (one paprika + tomatoes held; the completionist made the first-ever deliberate trip toward the paprika cluster and nav-looped six tiles short).

Costs. (1) The malformed-call attractor *mutated rather than died*: kwarg-in-key declined (154→86) but a new dominant form appeared — Python-call syntax inside the function tag, `<function=gather("Oak")>`: 0 (base) → 79 (r1) → **599** (r2) — plus corrupted closing tags. Abstention masking removes reinforcement but supplies no corrective pressure; the model's own malformed history compounds in-context. (2) The OPD models carry a higher total failure rate than base (base 8.6% / r1 16.9% / r2 16.7%), driven by a validation-failure class (≈0.1% at base → ≈11%) that the malformed calls produce; the game-state error class, by contrast, improved across rounds (8.5/5.5/5.9%). (3) Deaths rose monotonically (142/172/188), though 78% of r2 death positions are on the quest NPC's tile inside a level-45–54 aggro zone: substantially the *cost of attempting the quest* (turn-ins survive death; agents name the danger and proceed).

8 Round 3: counterfactual grading, full-ladder seeding, and a harness affordance

Round 3 targeted the two open wounds — the mutated malformed-call attractor and the stalled Herbalist stage 2 — with two weights-side moves and, ultimately, a harness one. **Counterfactual-canonicalized grading:** at parser-flagged records the teacher scores the student's emission under a *canonicalized* context s' — the system prompt's Python-style tool-doc literals rewritten to canonical wire syntax — while the student/behavior side keeps the real context s. Scoring the teacher under s' turns its clean-convention preference into a corrective negative advantage on the malformed tokens, rather than round-1's +0.09 copy-prior endorsement. This is a designed reward, not an unbiased reverse-KL estimate (student and teacher are scored under different contexts); we adopt it deliberately at exactly the spans where the visitation context poisons the grade. A pre-build *flip* *probe* set the scope empirically: canonicalizing the malformed *history* calls did nothing (a measured 0% flip), but rewriting the system prompt's Python-style tool-doc literals in the teacher's grading copy flipped 86% of flagged states at −1.21 nats — so s' rewrites the doc literals only. **Full-ladder seeding:** two seeded collections placed the 2B at every unsolved stage (Herbalist stage 2; Rick's fishing/cook/turn-in/door), all graded by the plain 4B. Build: 8,856 train / 1,040 held-out, 4 score failures; 365 counterfactually-graded records (zero masked — corrective grading replaced abstention). Gate PASS (+6.4% rKL, every verb).

The decisive negative result. The first eval was stopped at 35 minutes: counterfactual grading did *not* suppress the malformed *emission* — it regressed it, paralyzing two of three agents (57–76% of their actions were malformed calls the server drops, which fall to text, never execute, and the model re-emits in a spam loop). **Grader-endorsement-suppression, flip-probe-verified, did not transfer to student-emission-suppression after training** — the defect lives in generation, and a grading-context fix cannot reach it. The cure was a *harness affordance*: when the server drops a malformed call, the harness recovers the executable call (99.5% coverage), rewrites history to a clean canonical turn (severing the in-context copy prior at its source), executes it, and returns a loud format-correction note.

Result: 18/30 — the program's best, 6/10 **for every agent, and a clear step beyond base.** Every agent now completes two Core-3 chains unseeded — Foresting *and* Herbalist — where base completes one (Foresting, then stalls at the Herbalist wall). All three completed Herbalist 3/3 unseeded (round 2 had zero), breaking the stage-2 wall via three concrete differences from r2's stuck behavior: navigating to the *actual* paprika bush (deep in the hostile zone) rather than the prompt's stale coordinate; recovering from a one-tile navigation stall by re-targeting an adjacent tile (where r2 looped identically ∼16 times); and a tighter observe–act loop (half-length reasoning re-observes often enough to notice a frozen position) — with gathered items surviving death. The harness recovery worked and *self-corrected*: of 405 sessions that emitted a format note, every one has exactly one note and zero relapse (98.5% clean afterward); the malformation is a session-opening artifact the note durably fixes. Validation/format errors halved (10.7% → 5.1%); malformed emissions stayed bounded (335 vs r2's 685 — feedback breaks the compounding spiral). Reasoning compressed sharply against round 2: non-empty median 243 chars vs r2's 558, ∼30% near-empty turns (r2: 0%), revision-churn down (59% → 30%), and the dangling-</think> dialect abandoned (99% → 12%), with grounding preserved.

Decomposing the gains: weights are the competence engine over base. The full +6 over base (12→18) decomposes into +3 **from weights alone** (round two, 12→15, run *without* the harness affordance — the Herbalist wall falls 3/3 unseeded), +2 **from the harness recovery affordance**, and a residual +1 **from the round-three weight update that sits within the** n=1 **noise band** — it may be ≈0 stages; the round-three update's *reliable* payoff is speed, not stages. We isolate the harness/weights split *at the round-two-to-three step* with a controlled ablation: round-2 weights running with the recovery affordance on (otherwise the identical 6h protocol) already scores **17/30** (grinder 6, completionist 6, explorer 5) — so of the three stages from r2 to r3, ∼2 are the affordance and ∼1 the weight update (within the n=1 band). The cumulative and marginal pictures point opposite ways, and both are true: *over base, weights instilled the competence* (the +3 wall passage is weights-only, visible with seeding off at eval); *within the final step*, the harness did the stage work and the weight update's distinct payoff is *speed* — r3 reaches Herbalist stage 2 in ∼2.0h vs r2's ∼4.5h (∼2× faster), ∼40% more turns per hour (511 vs 366), the dividend of its shorter reasoning and fewer stalls. The slogan *harness → stages, weights → speed* describes the r2→r3 step only; over base the order reverses.

Rick's Roll stayed 0/4 **— a three-link execution failure, not missing intent.** The agents understood and attempted the full chain (52 navigations to Rick, 5 NPC-interaction attempts, 12 door attempts, all quoting the walkthrough). The chain dies at link one: the fishing pole is never re-equipped (Foresting's logs displaced it), so every shrimp gather returns nothing and cook/door/Rick never become reachable; `craft_item` was called *zero* times in all post-Herbalist sessions — exactly the cook-incompetent-teacher prediction (the 4B never cooks, so its seeded-state grades there were the flat KAT-agreement-trap signal). Seeded-state grading instilled the *declarative plan* but none of the *procedural gates* a 2B cannot clear. This is the clean separation round 3 establishes: **visitation + a competent teacher transfers competence** (Herbalist stage 2 broke); **a harness affordance dissolves what weights-side fixes cannot reach** (the format defect, after r1 created it, r2's masking contained it, and r3's grading regressed it); and **a teacher cannot grade in what it cannot do** (Rick's). The 18/30 arm is a weights + harness-recovery configuration, not pure weights — the data argues that is the correct framing, because the binding constraint was a model–environment interface failure.

9 What is and is not valid

Supported observations. (a) The r10 SFT arm fell below base in every one of seven exploratory runs (base $[7, 7, 7, 7]$, SFT $[3, 1, 2]$; unadjusted exact tests p=0.029 / 0.016). The aggregate tool-frequency match is a mechanism hypothesis, not a matched-state causal result. (b) In a small, duration-mismatched capacity sweep, larger models did not monotonically improve Core-3 score; this is scouting evidence, not a scaling conclusion. (c) Round-1 style-without-competence transfer (three independent measurement families agree: tool mix, tempo, per-verb KL signs). (d) The copy-prior mechanism (direct two-sided probe; three alternatives refuted by measurement). (e) The round-2 lift and its causal reading (pre-registered endpoint; unanimous across the only replicate structure we have). (f) The round-3 results: the Herbalist stage-2 break and its three-part mechanism (correct coordinate, stall-recovery, tight loop — all read directly from the transition sessions); the harness-recovery self-correction (405 sessions, exactly one note each, zero relapse — a counting fact, not an inference); the grader-vs-generation gap (the flip probe verified −1.21 nats of endorsement suppression, yet the stopped run shows the emission *regressed* — a falsifying observation); and the cook null (`craft_item` = 0, pole-equipped = 0, both across all agents — ground truth).

Faults and caveats, enumerated. *(1) The* 18/30 *arm is not pure-weights.* It ran with the harness tool-recovery affordance on; base/r1/r2 did not. The ablation (round-2 weights + recovery = 17/30) decomposes the round-2-to-round-3 gain as ∼2 stages from the affordance and ∼1 from the weight update, within the n=1 noise band; the round-three weight update's clean payoff is speed, not stages. A pure-weights 18/30 is not demonstrated — though the competence-over-base claim does not rest on it: round two's 15/30 wall passage is pure-weights (no affordance) and is the cleaner evidence. *(2)* n=1 *per arm*, agents are prompt variants not independent seeds; direction-consistency is suggestive, not significant. Stage counts are coarse (±2–3/30 historical band); error-rate and milestone analyses carry the statistical weight. The r10 negative is our only powered result; the beyond-base claim — even at 3/3 vs 0/3 on the wall — rests on prompt-variant arms that share a base policy and so are a partial (not independent) replicate, which is why we lean on the categorical wall passage rather than the stage delta. The obvious next validation is seed replication of the round-two passage, which we have not run. *(3) The malformed-emission counter is a lower bound.* The harness rewrites assistant content to canonical before logging, so the counter catches only malformed text that survived recovery (335 logged vs the authoritative 405 recovery-marker count); the true round-3 malformation count sits between the two. *(4) The mixed-context advantage is a designed reward, not an unbiased KL estimate*: the teacher is scored under the canonicalized context while the student is scored under the real one. CCOPD [33] uses the same move without a bias analysis, so we inherit that gap. *(5) Residual analyzer caveats*: quest-transition verb attribution reports the in-flight tool not the cause; `gather` item counts race the server and under-report successes; the eat_food error class conflates full-HP/slot-shift with validation empties. *(6) Costs are billing-verified* (`modal billing report`, as for r10's \$360): r1–r3 totaled \$160 incl. all debugging ($\sim$44% of one r10 SFT cycle), but the per-round split is the opposite of intuitive — round 1 $\approx$\$111 (dominated by a 3× broken-kernel $\sim$8h H100 retrain at $\sim$168 s/step), round 2 $\approx$\$33, round 3 $\approx$\$16. Once the fast GDN kernels were fixed (round 2) the marginal build+train of a round is $\sim$\$5–10; eval serving on L4 ($\sim$\$6–23/round) then dominates. No paid-API cost enters the loop — student rollouts and the 4B teacher are both self-hosted Qwen on Modal. *(7) The decode regime is non-thinking, by inheritance not design.* Every model ran the Qwen3.5 template's default closed-empty <think></think> prompt (the thinking-mode *sampling* preset sits on top of it), so reasoning surfaces as plain content; we did not deliberately choose thinking vs. non-thinking. It is held identical across base, teacher serve, teacher grading, and every student eval, so it is an internal-consistency property, not a confound on any comparison — whether thinking-on would change the results is untested.

Not claimed. No general statement about 2B capacity ceilings (Rick's failure is multi-causal — a displaced tool, a budget limit, and a teacher null — not a clean capacity result). No claim that grading-context fixes cure generation defects (the stopped run falsifies it). No claim that masking cures teacher-shared defects (the attractor mutated 154 → 86 on one form while a new form went 0 → 599). No cross-environment generality; findings are conditional on a deliberately strong scaffold whose transfer is untested.

10 Related work

Game-agent systems. Orak uses MCP as a plug-and-play interface across twelve games and includes a gameplay fine-tuning dataset [2]; MCP and structured game actions are therefore infrastructure here, not a novelty claim. BALROG evaluates long-horizon language and vision-language agents across games and documents a knowing–doing gap [3], while lmgame-Bench studies modular gaming harnesses and transfer from single-game training to unseen games and external planning tasks [4]. Unlike these benchmark-centered systems, our question is how a small policy's persistent environment-state visitation controls what dense teacher grading can transfer.

On-policy distillation. Reverse-KL on-policy distillation for LLMs was established by GKD [1] and popularized as a post-training recipe by Thinking Machines [5]; the 2026 wave maps its failure modes: trust-region restriction of unreliable teacher tokens [8], token-subset selection [10, 11], structural-token masking [12, 13], verifier-gated sign-consistency [14], trajectory-filter-then-soft-reweight [15], and the KL agreement trap, where low per-token KL on degraded prefixes is the failure signature [7]; DistIL proves reverse-KL-family objectives can raise probability on worse actions [9]. Our copy-prior finding is the adversarial sharpening of these: not absent corrective signal but a wrong-signed one, measured in a deployed agent. The visitation-coupling lens is formalized in States-not-Tokens [17], and our round-1 outcome — style-level convergence without capability gain — is its documented failure mode when the teacher's edge lies outside the student's visited states [27]; state-distribution control by resetting to hard/intermediate states is well-trodden in RL (reset-along-a-demo [19], Backplay [20], Go-Explore [21]) and appears LLM-side as branched rollouts from expert anchors [18], expert-state reset mixtures [22], resume-from-verified-prefix [23], and behavior-policy blending [24] — our contribution lifts this reverse curriculum into persistent typed environment state for on-policy distillation. Multi-turn agentic OPD at small scale documents per-turn KL escalation and curricula [25]; iterated rounds are the DAgger loop [26, 6]. The harness-over-capacity finding aligns with the 2026 system-scaling literature (interface adaptation over weights) [30, 31]. The r10 mechanism connects to imitation collapse / capability-suppression results in SFT [32].

Grading-context engineering (round 3). Scoring student tokens under a teacher context that differs from the generative context is an emerging family: CCOPD grades under a *canonical* clean context to fight self-anchored drift [33]; OPCD conditions the teacher on internalized context [34]; MixSD switches grading context per token [35]; CoD combines original and counterfactually-perturbed teacher contexts off-policy [36]. Our canonicalization of the teacher's grading context (the system-prompt tool-doc literals), and the finding that it suppresses *endorsement* without suppressing *generation*, are to our knowledge new; the mixed-context bias is uncharacterized in this family and we inherit that gap. **Privileged-information distillation** is the published route for Rick's Roll, where the same-family teacher is incompetent: a PI-conditioned teacher (verified or frontier-model trajectories in its context) scores the student's rollouts token-by-token [37, 38], with evidence-masked updates [39]; this enters as prompt text and so sidesteps the cross-vocabulary problem that sank our SFT line, and logit-free chunk-similarity distillation extends it to black-box teachers [40]. On the negative-gradient question, AOPD finds bare unlikelihood reinforcement ineffective and prefers corrective distribution-matching [41] (supporting our corrective-grade-over-mask choice), while AntiSD demonstrates a sign-reversed self-distillation term under an entropy gate [42]. The copy-prior mechanism's substrate is in-context copy bias and induction-head dynamics [28, 29].

11 Discussion and future work

Round 3 confirmed the predictions of round 2's failure analysis and sharpened the open agenda. What worked: full-ladder seeding broke the next teacher-competent wall (Herbalist stage 2, 18/30), and a harness recovery affordance with in-context correction dissolved the format defect that weights-side fixes (masking, then counterfactual grading) could not. What did not: a same-family teacher transferred zero of a behavior it never performs (cook) — though Rick's 0/4 is over-determined, gated at link 1 by a displaced fishing tool before the cook step is even reached. The round-4 agenda is now *evidenced*, not speculative: (i) **privileged-context grading** for Rick's — a Claude cook/door trajectory in the *teacher's* grading context ([37, 38]), the only published route that exploits cross-vocabulary trajectories without the token-level distillation that sank our SFT line; (ii) **two cheap harness fixes the logs demand** — auto-re-equip the displaced fishing pole (the link-1 gate) and a session note that carries route-progress for un-accepted target quests (the traverse-budget gate); (iii) promote tool-recovery to a *permanent* affordance (it is that load-bearing) while tracking the pure-weights defect rate separately, so the weights vs harness contributions stay disentangled. Methodologically, the round's sharpest lesson is that a verified *grading-side* probe is a hypothesis about *generation*, not a guarantee: the teacher's endorsement of malformed syntax dropped -1.21 nats under canonicalized grading, yet training on that signal regressed the emission rather than suppressing it. Endorsement-in-grading and suppression-in-generation are different quantities, and only an interface affordance acting at generation time closed the gap.

12 Conclusion

A 2B model could not be *imitated* into competence (r10: 3.5× regression); could be *scaffolded* to its family's frontier (12/30); moved onto its teacher's distribution without gaining a single stage (round 1: 12/30); *gained* competence when — and only when — the training distribution was made to visit the states where the teacher is competent (round 2: 15/30, the wall passed 3/3 unseeded where base passed 0/3); and reached the project's best, 18/30, completing two quest chains unseeded where base completes one (round 3). The clean claim is *better than base*: across the three rounds the 2B gained two walls base never clears, and the cleanest evidence is the weights-only round-two passage — we do not lean on nominally edging the 4B teacher (18 vs 17), since the 18/30 ran a recovery affordance the teacher did not. Three rounds, three levers, one monotone curve: *weights instilled the competence over base* (the walls fell on training, visible unseeded), a harness affordance dissolved a defect no weights update could reach, and the round-three weight refinement bought speed. And the ceiling is honest: Rick's 0/4 is over-determined — a displaced tool gates link 1, a multi-session traverse exceeds the budget, and a teacher cannot grade the cooking behind them — so it waits on cheap harness fixes *and* a teacher that can cook, not a fourth round of the same one.

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